

Macroeconomics I

Problem Set 1

1. **(Solow Model Basics).** Suppose a Solow model in discrete time. The population grows at rate n , while technology, A_t , grows at rate g . Let the $w = \partial F(K, AL)/\partial L$ and $r = \partial F(K, AL)/\partial K$.
 - (a) Show that the marginal product of labor, $w = A[f(\tilde{k}) - \tilde{k}f'(\tilde{k})]$, where the variables with tildes are per-capita and per efficient labor unit variables.
 - (b) Show that, if F has constant returns to scale, we have $wL + rK = F(K, AL)$.
 - (c) Suppose the production function is a Cobb-Douglas: $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$. Find the steady state variables $\tilde{k}_{ss}$, $\tilde{y}_{ss}$ and $\tilde{c}_{ss}$ as function of the parameters.
 - (d) Let the “golden rule” be the savings rate that maximizes consumption in the steady state. Find the golden rule of this economy.
2. **(Constant Relative Risk Aversion Utility Function).** The most commonly used instantaneous utility function in macroeconomic models is isoelastic utility (also known as CRRA utility function):

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma \geq 0 \text{ and } \sigma \neq 1.$$

For questions (c) onwards, also consider two periods with discount $0 < \beta < 1$:

$$U(c_1, c_2) = u(c_1) + \beta u(c_2),$$

- (a) Show that if $\sigma = 1$, then $u(c) = \ln c$ (hint: use L'Hôpital's rule).
- (b) Define $\sigma(c) = -\frac{u''(c)c}{u'(c)}$ as the Arrow-Pratt coefficient of relative risk aversion. Thus $\sigma(c)$ indicates the individual's attitude towards risk. Show that σ is constant.
- (c) Let the marginal rate of substitution between two periods be:

$$MRS(c_2, c_1) = \frac{\partial U / \partial c_2}{\partial U / \partial c_1}.$$

The function u is homothetic if $MRS(c_2, c_1) = MRS(\gamma c_2, \gamma c_1)$. Show that U is homothetic.

- (d) Define the intertemporal substitution elasticity as:¹

$$ies(c_t, c_{t+1}) = -\frac{d(c_{t+1}/c_t)}{dMRS(c_{t+1}, c_t)} \frac{MRS(c_{t+1}, c_t)}{c_{t+1}/c_t}$$

Show that $ies(c_1, c_2) = 1/\sigma$.²

¹The intertemporal substitution elasticity is defined as the percentage change in the consumption growth rate in response to a percentage change in the real interest rate. Note that the interest rate is the intertemporal price of consumption, and that at the optimum (Euler equation) it equals MRS .

²This relationship between the intertemporal substitution elasticity and the coefficient of risk aversion is not general. There are preferences in which the two variables are not related (for example, *Epstein–Zin preferences*).

- (e) **Bonus (not required to submit):** redo all exercises with a CARA (Constant Absolute Risk Aversion) utility: $u(c) = 1 - \exp -\sigma c$. Verify that certain conditions above are not satisfied.
3. **(Negishi's Method).** Consider a two-agent exchange economy. The utility function for agent $i = 1, 2$ is given by:

$$U(\{c_t\}_{t=0}^{\infty}) = \sum_{t=0}^{\infty} \beta^t \log c_t^i. \quad (1)$$

In each period, agents receive endowments e_t^i determined by the following rule:

$$e_t^1 = \begin{cases} \hat{e}, & \text{if } t \text{ is even;} \\ 0, & \text{if } t \text{ is odd;} \end{cases} \quad e_t^2 = \begin{cases} 0, & \text{if } t \text{ is even;} \\ \hat{e}, & \text{if } t \text{ is odd.} \end{cases} \quad (2)$$

The endowment can be transformed into a final consumption good at no cost: $c_t = \hat{e}$. The objective of the problem is to find the optimal allocations using the Negishi method.³

- (a) Let α^i be the weight given by the social planner to agent i (where $\sum_{i=1}^2 \alpha^i = 1$). Define and solve the social planner's problem for this economy. Interpret the first-order conditions. How does the planner choose to allocate the economy's resources (intertemporally and among agents)?
- (b) What is the set of Pareto efficient allocations for this economy?
- (c) We want to find the Pareto efficient allocation that decentralizes a competitive equilibrium. Determine that the price of a unit of consumption p_t is equal to the Lagrange multiplier of the economy's resource constraint (normalize $p_0 = 1$). Define a transfer function in terms of weights, $t^i(\alpha^i)$, where t is the transfer each agent needs to be able to purchase the efficient allocation given their endowment sequence $\{e^i\}_{t=0}^{\infty}$.
- (d) Briefly and intuitively explain why the transfer that implements the competitive equilibrium is equal to $t^i(\alpha) = 0$ for all i . Find the weight α that determines the competitive equilibrium.

³Hint: see section 2.2.4 of DK's notes.